# Investify


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# Students' Submission

**Unsigned working draft.** The submission statement and signatures require actual student review/confirmation before submission. The preparation date is not a confirmed assessment date.

We certify that this report is submitted as the FYP-1 mid-report for the project titled *Investify*.

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# Anti-Plagiarism Statement

**Unsigned working draft.** The team must verify sources, original authorship and disclosed tool-assisted work before making the statement below. No similarity check or institutional approval is claimed here.

The students certify that this report is their own work and that all sources used in preparing the report have been appropriately acknowledged and referenced.

# Authors' Declaration

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We declare that the work documented in this report represents the work completed by the project team up to the FYP-1 mid-report stage. Contributions, tools, external components, and relevant sources are disclosed where applicable.

# Abstract

Investify is an educational equity decision-support prototype that combines technical indicators, news sentiment and explanations checked against recorded evidence. Its verifier checks visible prose and JSON independently and supplies a deterministic fallback. A prototype risk questionnaire conditions a separate retrospective linear comparator.

A fresh NVDA experiment processes 1,509 price sessions and 6,768 revision-conservative news rows. The clean five-day model obtains macro-F1 0.2843 versus a stratified-random baseline of 0.3287. A deliberately centred indicator triggers the leakage canary. Both corrected calibration modes produce only Buy decisions on 186 evaluation sessions; improved Brier scores therefore do not establish useful classification.

Original CryptoBERT and Cardiff classifiers complete the same 24 frozen MSFT posts, scoring macro-F1 agreement of 0.2564 and 0.6016 against AI reference labels. The small, imbalanced reference has no independent human validation, and deployment selection remains open.

Recorded-model, calibration and native TreeSHAP replay pass. The interfaces expose evidence, failure cases and recorded comparisons. Six assistant-teacher explanations form a construction pilot; the planned Phi-3-mini student remains untrained. Broader generalisation, live LLM usefulness, agreed risk semantics, institutional completion and genuine team review remain outstanding.

# Acknowledgments

The team will finalize acknowledgments to reflect actual guidance, review and contributions. Supervisor identity follows the supplied project record. No additional participant, reviewer or contributor is invented in this working draft.

# Executive Summary

Investify separates educational equity predictions, recorded evidence, explanations and checking. A label alone does not communicate its inputs, uncertainty or limitations. The implementation exposes five technical and six news features, checks visible prose independently of JSON, rejects known contradictions and supplies a verified deterministic fallback. Prototype risk answers condition a separate retrospective linear blend; they do not personalise the saved classifier.

The preserved September experiments and 372 historical decision records remain unchanged. A fresh Alpaca experiment now restores a genuine input-to-model execution path on 1,509 NVDA price sessions and 6,768 revision-conservative news rows. Original FinBERT scores 5,748 unique headlines offline. Six leakage arms run at one- and five-day horizons with fixed model settings, shared row sets and six folds. The clean five-day macro-F1 is 0.2843 against a training-distribution random baseline of 0.3287; the centred-feature mutation has lift +0.2104. The weak clean result and deliberate failure are reported together.

Corrected frozen-classifier and refit-transfer calibration modes each emit 186 Buy decisions. Raw frozen decisions are Buy 40, Hold 51 and Sell 95; their fitted isotonic mappings make Buy largest on every evaluation row. Frozen Brier improves from 0.7929 to 0.6105 while macro-F1 falls from 0.3182 to 0.2238. A fixed comparator estimated only from the 181 available calibration labels has evaluation Brier 0.6095. Improved Brier against the raw model therefore does not demonstrate an advantage over a simple class-prior forecast. Model-byte, scaler, margin/probability, calibration, label-endpoint and native TreeSHAP replay pass. All 372 new complete-guided explanations verify separately from the historical records.

The social pilot retains 200 real MSFT posts, 117 usable AI references and 83 disclosed exclusions, with 93 development and 24 chronological held-out posts. Both original classifiers now complete the same 24 held-out posts without errors. CryptoBERT predicts only neutral and has macro-F1 agreement 0.2564; Cardiff has 0.6016. Exact agreement is 15/24 and 16/24. Cardiff is the exploratory benchmark leader, with only two negative reference examples. Publisher bytes, tokenizer/configuration identities and lossless original Cardiff serialization are verified. Its unused native pooler is retained with exact

classifier-logit parity. Independent human validation and deployment selection remain open.

The Streamlit Social Study view recomputes recorded candidate scores from receipt-bound output bundles; it performs no live model inference. The later guided-demo viewer/selector checks have 28 passes, including 24 AppTest cases. The interface now opens with a presenter route, fresh experiment defaults, profile examples and three decision stages. Browser visual review remains unavailable. Other dated scopes retain 179 scientific contract passes with six legacy-input skips, eight acquisition-admission checks, historical portable/HTTP checks and native loader/pooler fixtures. These scopes overlap and are not added into a single full-suite result. Prepared remote CI has not executed.

The current assistant teacher construction pilot contains six verified explanations and four training/two validation exports. Factual JSON supervision is assembled from original records; the exact serving revision is unavailable. Phi-3-mini remains the planned, untrained student, and Mistral is fallback. QLoRA belongs to the later milestone. No measured human-comprehension or training-efficacy claim follows from this pilot.

Remaining first-mid work includes final report/deck delivery and inspection, required institutional facts and applicable signatures, genuine member contributions and review/rehearsal, current repository evidence and agreed social/risk acceptance criteria. Broader prediction, social integration into the engine, live LLM evaluation, personalised modelling and deployment remain distinct later work. This report is an unsigned working draft.

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# List of Abbreviations

| | |
|---|---|
| FYP | Final Year Project |
| FR | Functional Requirement |
| NFR | Non-Functional Requirement |
| API | Application Programming Interface |
| CI | Continuous Integration |
| CD | Continuous Delivery / Deployment |
| CV | Cross-Validation |
| F1 | Harmonic Mean of Precision and Recall |
| FNSPID | Financial News and Stock Price Integration Dataset |
| HTTP | Hypertext Transfer Protocol |
| JSON | JavaScript Object Notation |
| LLM | Large Language Model |
| MACD | Moving Average Convergence Divergence |
| NYSE | New York Stock Exchange |
| OBV | On-Balance Volume |
| PoC | Proof of Concept |
| QA | Quality Assurance |
| QLoRA | Quantized Low-Rank Adaptation |
| RSI | Relative Strength Index |
| SHAP | Shapley Additive Explanations |
| SMA | Simple Moving Average |
| UTC | Coordinated Universal Time |

# Chapter 1

# Introduction and Project Vision

## 1.1 Project Overview

Investify (F26-003-Investify) is an educational investment advisory project for retail investors. The planned system combines technical indicators, news and social sentiment, and risk information to present a one-week Buy, Hold, or Sell view with understandable, checkable evidence. The proposal was approved with minor modifications concerning contribution clarity, the language-model role, evaluation, transparency, and domain understanding.

## 1.2 Problem Statement

A retail investor evaluating a company must interpret historical price indicators and a large volume of news and social commentary. A decision label alone does not explain which signals influenced it, how conflicting evidence was handled, how uncertain it is, or whether an action fits the investor's risk constraints. A fluent explanation can misstate recorded values or attributions, while an evaluation using information unavailable at decision time can overstate model quality.

Investify addresses these issues through a temporally controlled evaluation pipeline and explanations tied to recorded inputs and model attributions. Its outputs are educational and advisory; predictive usefulness must be established against explicit comparators, and the system does not execute transactions.

## 1.3 Problem Context and Motivation

The baseline contains a historical NVDA experiment using technical indicators and financial-news sentiment. It found no demonstrated predictive signal, and its calibrated evaluation records all select Buy. Explaining those records makes the computation and its limitations inspectable; it does not establish a useful trading signal. Financial-text classification supplies evidence about text tone rather than the truth of a headline or a promised return [1].

## 1.4 Project Vision

The user can inspect a view's evidence, understand its uncertainty and coverage, and ask questions without receiving invented values or unsupported causes. The LLM communicates the recorded decision. A separate claim verifier and deterministic fallback govern supported factual statements.

## 1.5 Project Objectives

Objectives cover point-in-time alignment, engine/baseline comparison, equity social-model transfer and news/social disagreement, capacity/tolerance reconciliation, verified explanations and user understanding, and a reproducible measured system. Implemented components, measured findings, and planned objectives are reported separately.

## 1.6 Project Scope

The current prototype covers preserved NVDA artifacts plus a freshly executed Alpaca news/price experiment at one- and five-session horizons. A separate original-classifier selection study uses frozen MSFT social posts. Full scope targets a small equity universe, a one-week user horizon, social sentiment, genuine risk conditioning, and a verified LLM. Price-only continuously traded assets provide a later calendar experiment. Brokerage integration and trading execution are excluded.

## 1.7 Project Modules

Modules cover ingestion/storage, calendar/as-of alignment, feature/sentiment construction, evaluation/controls, prediction/calibration/attribution, risk profiling, explanations/verification, and interface/deployment. The new increment adds complete evidence context at the explanation/interface boundary.

## 1.8 Stakeholders and User Classes

The primary user seeks educational analysis. The operator maintains data/runtime health. The supervisor evaluates scope and evidence. Each team member must explain and modify their actual work and disclose external assistance and components.

## 1.9 Success Criteria

Criteria include causal alignment checks, injected faults, held-out predictive/calibration metrics, explanation measures, interpretation tasks, and runtime/installability checks. Existing explanation targets are numeric fidelity at least 0.98, directional consistency at least 0.95, recall at least 0.80, and unverifiable rate at most 0.15. Planned bounds are not measurements.

## 1.10 Constraints and Assumptions

The original September training store/raw corpora remain unavailable, and fresh upstream changes are unverified. A new Alpaca scientific run and the original social-candidate comparison now execute with separate identities. Live LLM/student inference and comprehension evaluation remain pending. Six recorded teacher examples demonstrate construction under a bounded contract. The prototype questionnaire affects a separate technical/news comparator; it does not condition the saved XGBoost predictions or implement the proposal's validated risk instrument. Context/verification checks can run without the training stack.

## 1.11 Ethical, Legal and DEI Considerations

The interface must explain data age, coverage, uncertainty, actual profile effect, and failure states. An AI explanation is not a guaranteed prediction [2]. Personal data, annotations, and acceptance work require appropriate privacy, licensing, consent, and truthful records. The advisory boundary must remain clear throughout the journey.

# Chapter 2

# Research and Existing Solutions

## 2.1 Research Method / Approach

This preliminary review uses primary research abstracts, the actual sentiment checkpoint's model card, official exchange/API documentation, and the preserved project artifacts. The comparison is focused on sentiment inputs, time alignment, attribution and explanation consistency. It is not a completed systematic literature review or evidence that no other product provides similar features.

## 2.2 Existing Solutions / Products

Araci's FinBERT work investigates domain-adapted language models for financial sentiment [3]. The implemented checkpoint, `ProsusAI/finbert`, outputs positive/neutral/negative sentiment scores and is documented separately by its provider [1]. Investify transforms these outputs into historical news features; their availability alone does not establish return predictability or social-domain transfer.

FNSPID combines financial news with stock-price data for forecasting research [4]. The project's preserved subset probe concerns the actual selected files/tickers and timestamp precision. It must not be generalised into a claim that every record in the published dataset has the same limitation. Supplemental Alpaca/Benzinga results remain a separate experiment.

Tree-based SHAP work provides consistent local feature attributions [5]. Investify uses stored predicted-class attributions to explain raw model scores. The verifier and educational display are downstream additions, not inventions of TreeSHAP or guarantees of economic causation.

Parcalabescu and Frank distinguish output-level self-consistency tests from stronger

claims about a language model's internal explanation faithfulness [6]. This motivates a precise boundary: Investify checks supported statements against an external decision record and a maintained glossary. It does not prove all natural-language meaning or the LLM's internal reasoning faithful.

## 2.3 Comparison of Existing Solutions

| Work/component | Relevant capability | Investify's use or evaluation question |
|---|---|---|
| FinBERT/checkpoint | Financial-text sentiment classification | Historical news features; equity social transfer still needs a labelled study |
| FNSPID | Integrated news/price research data | Audit subset coverage, publication precision and calendar alignment |
| TreeSHAP | Local attribution for tree models | Validate additivity and preserve raw-score units in explanations |
| Natural-language consistency research | Limits of explanation faithfulness tests | Define record-consistency claims and test concrete corruptions |
| Numeric deterministic template | Reproducible recorded facts | Initial explanation comparator and verified fallback |
| Definition-enhanced template | Same facts with maintained feature meanings | Fair comparator for any claimed LLM comprehension benefit |
| Local constrained LLM pilot | Structured candidate generation | Adapter tested; actual model outputs/evaluation unavailable |

## 2.4 Limitations and Gaps

The engineering gap demonstrated locally is that a correct JSON payload could hide an incorrect displayed explanation. Concrete regressions establish this defect and its correction. A broader research novelty claim requires comparison with the closest record-grounded generation and checking approaches. Adding an LLM or SHAP alone is insufficient. Human understanding, unrestricted language verification, social transfer and useful

stock prediction remain unestablished.

## 2.5 Closest Factuality and Annotation Work

FActScore decomposes generated content into atomic facts and assesses support against a knowledge source, including a retrieval/model-based estimator [7]. Its open-domain factual-precision setting motivates claim-level accounting. Investify instead limits accepted language to a maintained grammar and external numerical record, and checks required visible evidence separately from factual precision. Accurate included facts do not establish that required evidence was included. This is a scoped engineering distinction; a new general factuality method or empirical advantage over FActScore is not demonstrated.

RARR researches attribution and revises unsupported generated content while preserving the original output as much as possible [8]. Investify's correction policy rejects a failed bounded-contract attempt, optionally retries, and emits a checked deterministic guide. The planned pilot supplies that complete guide to the model. A meaningful educational advantage therefore requires further evaluation beyond successful reference reproduction.

Gilardi and colleagues study ChatGPT text annotation on relevance, topic, stance and framing tasks [9]. That research motivates investigating AI references, but does not validate the current financial labels, unknown exact serving revision, subjective exclusions or historical MSFT cohort. Candidate studies must report agreement with one AI reference pass and retain the independent-human-validation limitation.

## 2.6 Official Publisher Model Policy

The current assistant is the active explanation teacher under user direction; Phi-3-mini remains the planned original Microsoft student, with its exact checkpoint pending. Mistral is a fallback and has not been called. Six controlled-prose teacher examples are recorded, with factual JSON assembled from the original records as disclosed hybrid supervision. Exact teacher serving revision, student inference, training benefit and user comprehension remain unmeasured.

Local model baselines must use original publisher weights or publisher-issued quantizations, excluding third-party fine-tunes and behaviour-modified variants. Qwen2.5-7B-Instruct-GGUF and Qwen3-8B-GGUF are earlier inactive optional local configurations, not the selected teacher or student. Their catalog declares exact revisions and every weight-shard SHA-256/size; actual weight verification is required before those optional studies. Acquisition was refused and no replacement inference score exists. Frozen AI

references and historical classifier artifacts remain separate. The project's later QLoRA adapter must be versioned and disclosed separately from unchanged student weights.

## 2.7 Proposed Solution

Investify combines temporal evaluation controls with explanation claims traceable to recorded inputs/attributions. Visible text and JSON are checked independently, known contradictions fail, and unsupported/failed generations select a verified deterministic explanation. The local pilot includes exact maintained definitions and compares numeric template, definition-enhanced template and real generations on a frozen chronological selection. No quality score is reported if the provider returns no completion.

## 2.8 Complex Computing Problem

The problem combines asynchronous text and trading calendars, overlapping prediction horizons, fold-local processing, model calibration/attribution, and generation whose displayed claims must remain consistent with external numerical facts. The interfaces have competing constraints: understandable explanations need adequate coverage, while checking must avoid accepting unsupported claims or rejecting all useful content. Institutional mapping to the required complex-computing-problem criteria still requires supervisor/team review.

## 2.9 Technical Challenges

Publication precision, session/half-day boundaries, missing data, calibration collapse and raw-score interpretation complicate the baseline. Original September inputs remain absent; separate genuine Alpaca inputs and a scientific Python 3.11 runtime now support executed experiments. Fresh default LM Studio/Ollama probes return no model identifier. The inspected model directory names an old C: target, while E: contains application/backends and a bundled embedding model but no discovered generative weights. Global configuration is unchanged; these observations do not establish all failure causes or generation/deployment completion.

The social-model study distinguishes post sentiment from return prediction. CryptoBERT's provider documents crypto-social training, bearish/neutral/bullish class indices and a 128-token training length [10]; a generic Twitter alternative documents different sentiment names and mention/URL preprocessing [11]. Equity-domain suitability therefore needs the actual labelled study, not a model-name preference. The local-only run-

ner records explicit mappings, token/truncation information and failed outcomes; missing/randomly initialized classifier heads are refused. Earlier tests use fixture backends; the later original-publisher comparison now executes both classifiers with raw-logit and byte-bound receipts.

FinLin provides a multi-source entity-level research design, with individual annotations in its described schema [12, 13]. The inspected repository contains descriptions/figures rather than the corpus; no access request has been sent. The comprehensive StockTwits project describes public-access joined message/metadata tables and user sentiment tags [14]. Such tags do not provide new independent human labels or a neutral class. Corpus access, permitted use, published-score handling and pretraining overlap require explicit review. A chronological benchmark split alone does not prove absence from model pretraining.

## 2.10 Research Findings and Design Decisions

Stocks represent ownership interests, and price movement can reflect company-specific or wider market events [15]. The prototype's technical/news features do not provide a full valuation or represent every fundamental/macroeconomic driver. A worked explanation must distinguish what its actual inputs measure from unsupported claims about why a market price changed. Corporate-action adjustment, broader drivers and changing regimes still require dedicated review.

NVIDIA's official 2021 FAQ documents a four-for-one split and trading on the new basis from 20 July [16]. As a worked arithmetic illustration, one share at 100 units becoming four shares at 25 units preserves 100 units of value; treating the per-share change as a 75% investment loss would omit the share-count change. These numbers illustrate arithmetic and are not observed NVDA prices. Yahoo describes adjusted close as accounting for splits and distributions [17]. The current ingest disables yfinance's client adjustment and omits action columns [18]; that setting alone does not verify provider Close/Volume are as-traded. The raw snapshot, action history and volume basis require audit before any adjustment or performance claim. This issue has not been established as a cause of constant-Buy output.

Official exchange hours include regular and early-close schedules [19]; alignment must therefore use an explicit decision-time contract and exchange calendar rather than calendar-date equality. The calendar cutoff/proposal reconciliation remains open. Model-card sentiment outputs motivate a news-specific pipeline and separate equity-social study. Attribution theory motivates additivity checks and clear raw-score units. Explanation-consistency literature motivates bounded, explicit claims rather than a claim of universal semantic faithfulness. Local structured-output documentation supports JSON-schema

requests, but schema conformance does not replace factual checks [20].

## 2.11 Actual AI Reference Pilot

The author-released StockNet corpus provides original social-post content, identifiers and timestamps [21, 22]. The first NVDA sampling declaration could not execute because that ticker is absent from the released directory list. MSFT was declared before content acquisition. The actual AI reference pilot contains 200 real MSFT tweets: 117 usable labels (31 positive, 67 neutral, 19 negative) and 83 explicit exclusions. Its frozen split is 93 development/24 held out, with only two negative held-out references. One session pass is recorded with full prompt/input/raw-output/timing and declared model-identity provenance; agreement/kappa are null and independent human validation is absent. Both original candidates now complete the same 24 held-out posts; their exploratory agreement results are reported in Chapter 7. Public historical posts may occur in annotation or candidate pretraining. Exact deduplication leaves near-duplicates and retweets. Whole-post sentiment candidates and target-equity references differ in scope. These limits prevent an independently human-validated accuracy or NVDA transfer claim. The retained MIT repository notice does not authenticate third-party tweet rights; this workspace has not publicly redistributed the corpus.

# Chapter 3

# Proposal Feedback and Iteration Plan

## 3.1 Proposal Feedback Summary

Project F26-003-Investify was approved with minor modifications. The panel requested clearer contributions beyond leakage, the LLM's user benefit, technical/sentiment explanations, refined problem/model descriptions, explicit metrics, an improved PoC, user transparency, and deeper market understanding.

## 3.2 Proposal Action Register

The working register preserves the full supplied wording. These identifiers organize project actions; owners and dates require team agreement.

| ID | Required response | Current evidence/status |
|---|---|---|
| PF-01 | Clarify contribution | Revised statement drafted; related-work comparison pending. |
| PF-02 | Explain LLM benefit | Six actual recorded teacher examples; live inference and demonstrated user benefit pending. |
| PF-03 | Explain technical/sentiment evidence | Complete technical/news context and recorded social comparison; live LLM benefit pending. |
| PF-04 | Refine problem statement | Working revision drafted. |

| | | |
|---|---|---|
| PF-05 | Clarify pretrained models | Original classifiers compared; deployment criteria and live LLM evaluation pending. |
| PF-06 | Identify metrics | Matrix and current checks documented. |
| PF-07 | Improve PoC beyond leakage | Corrected verifier/local HTTP demo checked; browser and supervisor demonstration pending. |
| PF-08 | Transparency throughout journey | Data, coverage, and risk limitations exposed; complete failure paths pending. |
| PF-09 | Market understanding | Referenced notes begun; broader research pending. |

## 3.3 Iteration Strategy

Each increment supplies a working path, tests, and evidence. The independent code copy retains Git history and uses base commit 1dc24e0. Existing documents and the original checkout remain protected. Upstream access failure is recorded rather than taken to mean no new work exists.

## 3.4 Iteration 1 Plan

Original commitments cover ingestion, calendar mapping, indicators, news sentiment, social-model selection, the linear baseline, leakage tests/mutation detection, an interface slice, and requirements/design. The first new increment establishes artifact validation and complete evidence presentation. Next work connects and evaluates a real prompted explainer, then addresses failure behavior and risk semantics.

## 3.5 Iteration 2 Plan

Later work follows the proposal's engine, divergence, deployment, grammar, language-model comparison, runtime, and acceptance milestones. The feedback does not itself require the entire QLoRA programme at this mid.

## 3.6 Work Breakdown and Responsibilities

Proposal and repo records disagree about harness ownership. The team must reconcile actual contributors and defence responsibilities. Historical contribution is not reassigned or manufactured in this report.

## 3.7 Progress Against Plan

The earlier 12 October selected suite has 212 passes and no skips, including 20 actual Streamlit cases. All 372 records/complete-guided templates, 62 live HTTP checks and 228 separate comparator arithmetic probes validate. A prototype profile/blend, restored viewer, social-study validator/evaluator/view and offline classifier runner are implemented. Six actual assistant-teacher explanations pass the bounded verifier; this does not establish added explanatory content or user benefit. The source-handover verification is refreshed after each archive change and recorded separately; it excludes external scientific/UI packages. Fresh science and original sentiment-candidate results are now recorded. Live student/LLM inference, final visual/institutional review and remote CI remain pending.

## 3.8 First-Mid Boundary

The supplied rubric evaluates the planned milestone, a meaningful live core module, requirements/design, testing, process evidence, report and student ownership. It sets no fixed whole-project implementation percentage. The preserved proposal places the scored prompted baseline and filtered training set at Mid-2 (week 18), and QLoRA with the three-way comparison at Final-2 (week 24). The first-mid commitments remain distinct from cached-record replay. Fresh ingestion-to-baseline/leakage execution and the original social comparison now provide actual evidence; unresolved scope, selection and review issues still need explicit closure.

## 3.9 Delays, Deviations and Recovery Plan

Repository access, original September bytes and exact-lock restoration still constrain historical reproduction; the new separately identified experiments execute. The portable artifact path permits tested explanation work while those inputs are restored. The assessment date is unconfirmed. The full mid package is estimated at 45–75 person-hours, subject to data and human-study access.

# Chapter 4

# Software Requirements

## 4.1 User Interactions / Use Cases

The increment lets a reviewer select a historical corpus/session, inspect all technical/news evidence, check explanations, inspect rejection/fallback, complete an explicit prototype risk profile, and compare a separate profile-dependent linear blend. It identifies absent social features, historical data, unchanged classifier probabilities and constant-Buy behaviour. The final retail-user journey remains broader than this demonstration.

## 4.2 Use Case Diagram

The implemented interactions are Select Historical Decision, Inspect Evidence, Check Explanation, Inspect Rejection/Fallback, Calculate Prototype Profile, Compare Separate Blend, and Inspect Social Study Evidence. The diagram represents the current reviewer/operator journey, not a complete approved retail-product design.

## 4.3 Detailed Use Cases

For Check Explanation, the precondition is an available, validated record. The user submits visible text and optional JSON claims. Both are checked against the record. A contradiction, missing essential display claim, malformed payload, or unsupported sentence in strict mode causes rejection and a verified deterministic fallback. An unavailable record produces an explicit error instead of a substituted decision.

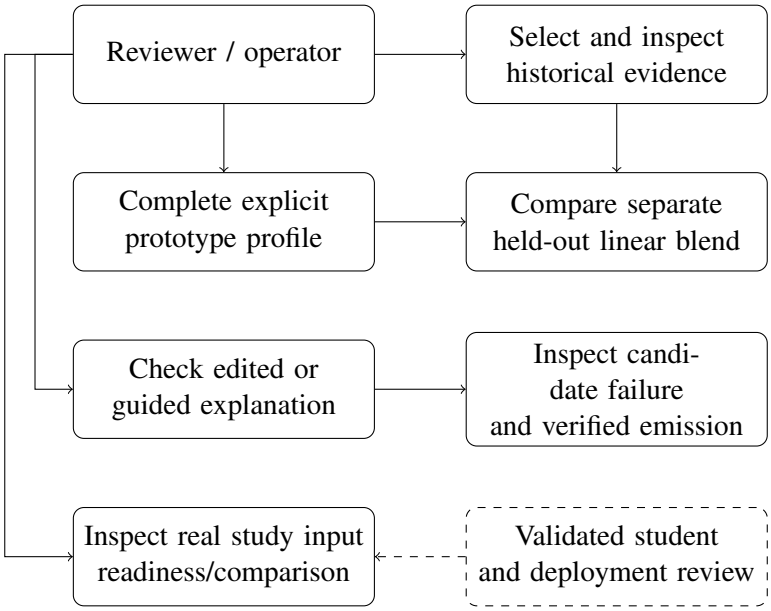



Figure 4.1: Implemented mid-evaluation reviewer/operator interactions and explicit study dependencies.

## 4.4 Functional Requirements

These identifiers document this increment; they do not replace the full proposal baseline or imply supervisor approval of a scope change.

| Identifier | Requirement | Status |
|---|---|---|
| MID-FR-01 | Load real saved decisions; disclose unavailable/invalid inputs. | Implemented |
| MID-FR-02 | Show all technical/news features with values, attributions, meanings, and limitations. | Implemented |
| MID-FR-03 | Check visible prose and JSON independently against the record. | Implemented |
| MID-FR-04 | Reject known contradictions and missing essential claims; verify the fallback. | Implemented |
| MID-FR-05 | Retain the candidate's rejection report separately from the fallback report. | Implemented |
| MID-FR-06 | Reject unparsed sentences in strict mode. | Implemented |
| MID-FR-07 | Disable scientific re-runs when input files are absent, empty, unreadable or structurally invalid, or runtime modules are missing. | Structural preflight tested |

| | | |
|---|---|---|
| MID-FR-08 | Request only an explicitly configured local generator; bound retries/timeouts and disclose fallback. | Adapter tested; real model pending |
| MID-FR-09 | Check complete maintained feature definitions in both visible text and JSON. | Implemented |
| MID-FR-10 | Preserve raw/failed attempts and frozen evaluation selection; report no model-quality score without completions. | Implemented; plan recorded |
| MID-FR-11 | Check every feature's value, SHAP magnitude/direction/target, family totals and attribution scope in both guided representations. | Implemented |
| MID-FR-12 | Require all profile answers; expose capacity/tolerance reconciliation and matched/binding project caps. | Implemented prototype |
| MID-FR-13 | Fit comparator scaling on earlier development rows; refuse development-session decisions and missing technical inputs. | Implemented |
| MID-FR-14 | Show the profile-dependent blend beside a fixed reference, preserving saved classifier probabilities and disclosing unvalidated assumptions. | Implemented prototype |
| MID-FR-15 | Validate social-post provenance, declared AI or human initial labels and annotation provenance, adjudication/exclusions and a frozen chronological study plan. | Frozen AI reference; paired runs recorded |
| MID-FR-16 | Require matching code/data/checkpoint/preprocessing and complete paired held-out outputs for sentiment comparison; preserve failures/raw runs. | Actual paired study; review pending |
| MID-FR-17 | Run a declared social classifier only from existing local weights with explicit class/text/token mapping; retain logits, truncation and failures. | Both original classifiers executed |
| MID-FR-18 | Retain actual teacher prose and provenance; verify it independently of assembled JSON; exclude test dates and export training/validation separately. | Six-example construction pilot; student pending |

## 4.5 Non-Functional Requirements

The original quantitative NFRs remain separate from this increment's scoped engineering contracts. None of these planned performance bounds is accepted from template, fixture, HTTP health or import tests alone.

| ID | Preserved original acceptance target | Current status |
|---|---|---|
| NFR-1 | Feature assembly, inference, TreeSHAP and verification below 800 ms at p95; target VM, warm cache; excludes generation. | Unmeasured |
| NFR-2 | Sentiment throughput at least 150 documents/s on 16 GB GPU, fp16 and 128-token input. | Unmeasured |
| NFR-3 | TreeSHAP below 50 ms on CPU. | Unmeasured |
| NFR-4 | Peak resident inference memory below 4 GB on the target VM. | Unmeasured |
| NFR-5 | QLoRA within four GPU hours on the 16 GB envelope; later milestone. | Unexecuted |
| NFR-6 | Explanation including verification below five seconds; target VM, CPU. | Unmeasured; reconcile |
| NFR-7 | Nightly incremental ingest and scoring over the full universe below 30 minutes; GPU and local store. | Unexecuted |
| NFR-8 | Full purged walk-forward sweep and hyperparameter search below 90 minutes on CPU. | Unmeasured |

The local adapter's 20-second default socket timeout, configurable upper bound and two attempts do not establish a five-second whole-request deadline. Its bounded-failure behaviour is tested separately; NFR-6 is pending reconciliation and measurement. The inspected GPU has 10,240 MiB, so the proposal's 16 GB targets are not assumed satisfied. The separate recovered classifier runtime is CPU-only and is not a QLoRA environment.

| Scoped ID | Testable increment acceptance | Evidence/status |
|---|---|---|
| MID-NFR-01 | Preserve rejected-candidate reporting and verify fallback independently. | Verifier, emission and HTTP regressions |
| MID-NFR-02 | Show explicit unavailable-data/refusal states without fabricated decisions or experiment results. | Record, readiness and widget tests |

| | | |
|---|---|---|
| MID-NFR-03 | Local generator requests use loopback, bounded attempts/socket timeouts and retain failures. | Transport/adapter fixtures; whole deadline unmeasured |
| MID-NFR-04 | Expose only declared local routes, reject malformed/nonfinite/oversized requests and redact credentials. | HTTP/transport tests; prototype scope |
| MID-NFR-05 | Git-free delivery verifies included inventory and reports uncommitted identity explicitly. | Source/handover tests and archive receipts |
| MID-NFR-06 | A visible verification report must match the current record, text, JSON-inclusion and strict setting; obsolete record, source, profile and result requests cannot publish or release a newer request. | Eight pure-state assertions and twelve controlled Node handler cases; browser QA pending |

These contracts do not replace the original NFRs or certify participant comprehension, deployment security or scientific accuracy.

### 4.5.1 Reliability

Missing inputs do not produce invented decisions. A template failing its own verifier raises an error. The portable server calls the real pipeline rather than simulating generation.

### 4.5.2 Usability

Candidate and emitted text are labelled separately. Feature meanings, raw-score attribution units, and unsupported capabilities are disclosed. Visual QA and participant comprehension checks remain pending.

### 4.5.3 Performance

Interactions perform no training or corpus scoring. HTTP duration is recorded in the live evidence JSON; it is not a browser-rendering or full-system latency measurement.

### 4.5.4 Security

The portable server binds to loopback and serves a single page, one explicit state script and declared APIs. It does not expose arbitrary project files or provide brokerage access, authentication, or production deployment.

## 4.6 Acceptance Criteria

Both corpora retain their recorded facts and accepted templates. Correct JSON cannot hide wrong visible prose; correct prose cannot hide wrong JSON. Rejection yields the verified template and retained failed report. Missing inputs are disclosed without synthetic replacement. Strict mode rejects unsupported sentences while preserving correct templates.

## 4.7 Requirements Traceability

The appendix and the machine-checked working traceability matrix map these 18 identifiers to exact source symbols and named executed tests. The check establishes anchor existence and execution, not a coverage percentage or institutional acceptance. Full proposal requirements still need reconciliation with agreed owners, approved iteration dates, social-study evidence, risk semantics, LLM evaluation, and deployment.

# Chapter 5

# System Design

## 5.1 System Architecture

The recorded assistant teacher pipeline is separate from the user-facing student inference path. Six actual controlled-prose explanations have been checked and exported; original-student inference and QLoRA remain pending. No extra Qwen teacher download is required. The factual JSON target is assembled from original records, explicitly as hybrid supervision rather than a teacher-generated payload. The existing historical pipeline acquires and aligns prices/news, derives features, evaluates fixed models, and stores decisions/attributions. This increment operates downstream of those records and adds complete context, independent prose/payload checks, and a live local verification interface.

## 5.2 Architectural Components

The validating decision schema supplies facts. The context builder adds reviewed definitions without changing predictions. The template renders supported sentences. The verifier checks both visible and structured claims. The emission pipeline accepts a candidate or verifies a fallback while retaining its rejection report. Streamlit and the portable HTTP interface expose these components.

## 5.3 Interfaces and Dependencies

The portable interface uses Python 3.11's standard library and explicit catalogue, record, explanation, profile-contract and comparator APIs. Streamlit uses the separately frozen, restored viewer runtime. Historical re-runs require their original price, news

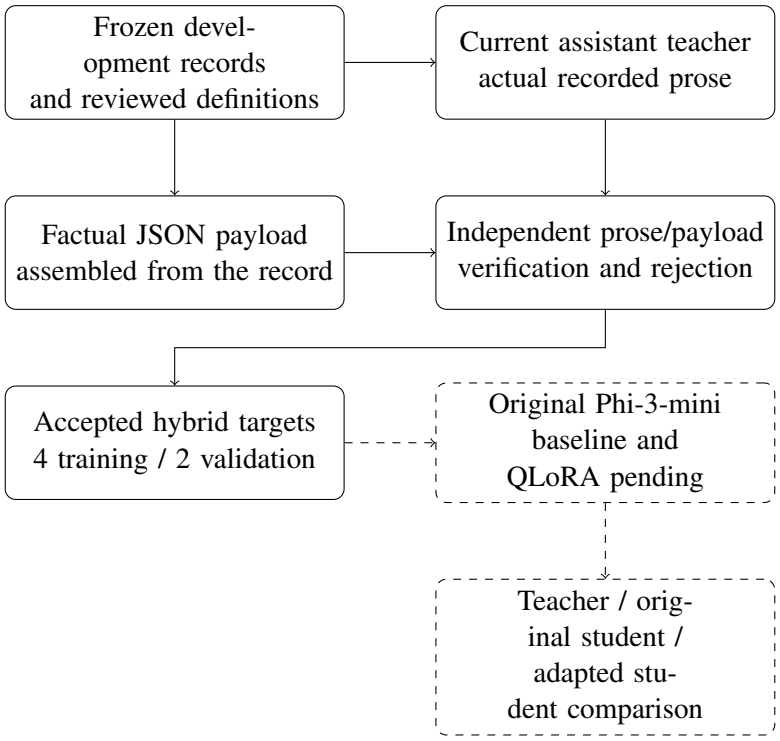



Figure 5.1: Actual teacher construction and verification, with pending Phi-3-mini training and comparison shown dashed. Both corpora share date groups and held-out test records are excluded.

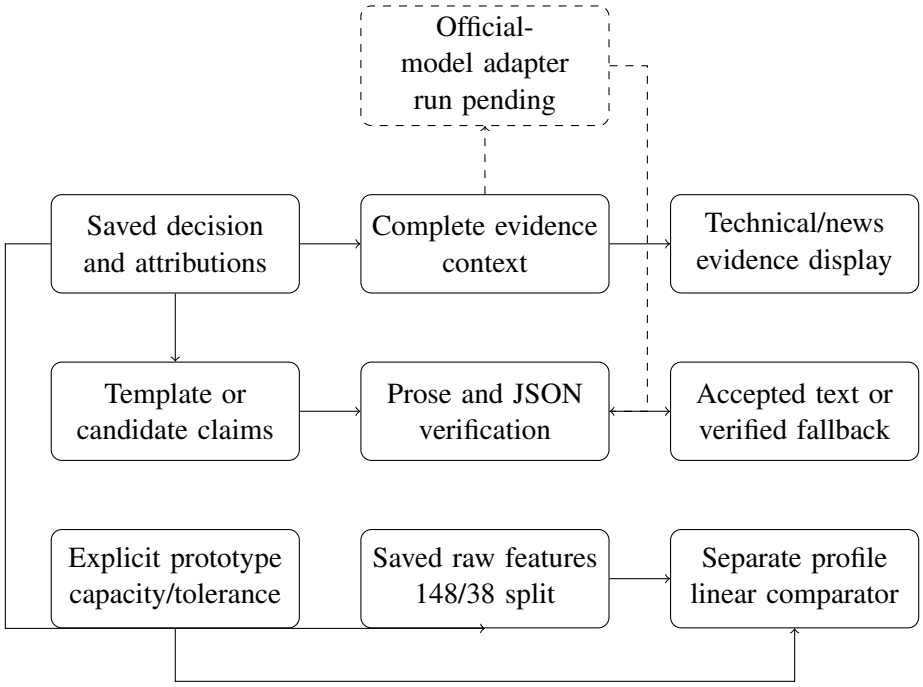



Figure 5.2: Explanation architecture with the implemented adapter and a pending official-publisher model run.

and sentiment-cache inputs. The fresh Alpaca experiment uses its separately restored scientific runtime and verified input identities; the viewer runtime remains separate.

| Interface | Request / response contract | Failure contract |
|---|---|---|
| GET /api/catalog | Available corpora/sessions plus unavailable and comparator-unavailable entries. | Partial availability is disclosed. |
| GET /api/record | Corpus/session query returns authoritative context and numeric template. | Unknown/unavailable selection: 400, no substituted record. |
| POST /api/verify | Text, optional payload and strict flag; candidate/emitted reports and replacement flag. | Factual rejection: 200 with candidate false, emitted true and replacement true. |
| POST /api/explain | Corpus/session; complete guided output, status/reason and retained attempts. | Unconfigured model: labelled guide, zero model calls. |
| GET /api/profile-contract | Five capacity and three tolerance controls, with no assumed answers. | No profile answers inferred. |
| POST /api/risk-comparator | Exactly corpus/session/capacity/tolerance; separate profile/fixed blend and audit. | Incomplete, development or unavailable request: 400. |
| GET /health | Process/data readiness and source hashes captured when the server starts. | HTTP 200 does not imply scientific or model readiness. |

Recognized POST transport failures are 415 for a non-JSON content type, 413 for invalid/empty or over-131,072-byte declared bodies, and 400 for malformed/nonfinite JSON or invalid top-level requests. Undeclared routes return 404. A syntactically valid but factually wrong candidate selects verified fallback rather than becoming a transport error. Numeric verification fallback and the complete guided explanation comparator have different coverage contracts.

For a valid request, the interface selects a validated record, builds evidence, submits current text/settings, verifies prose and JSON independently, and displays the accepted candidate/report. For rejection, the same path records candidate failures, verifies a numeric template independently and displays the fallback with the failed candidate report retained. Client edits invalidate the displayed verification report while preserving independent guide/profile output. Request ownership prevents obsolete catalogue, record,

profile, guided and comparator responses from publishing or releasing a newer request's controls. An accepted empty catalogue clears the current record/reports. Source refresh preserves an available selection; separate record and profile retries recover failed requests without assuming answers or duplicating controls. Twelve controlled Node cases execute the real inline handlers with a limited DOM/deferred-fetch fixture, separately from eight pure-state assertions. They make no live server, model or browser request; browser visual review remains blocked.

## 5.4 Architecture Decisions and Rationale

The record is authoritative over both representations. Coverage comes from visible text. Numeric and directional scores use the worse result of prose and JSON; payload duplication cannot dilute a display error. Any known incorrect claim is a hard failure. Strict mode rejects unparsed sentences and does not claim to verify arbitrary market prose.

## 5.5 Design Evolution

The original implementation intentionally preferred payload claims. A new regression showed that wrong displayed confidence could pass beside correct JSON. The corrected contract checks both, adds feature-value/attribution-target validation, requires essential display claims, and retains the failed report. Prediction outputs and historical experimental results are unchanged.

## 5.6 Domain Model

A corpus contains session-keyed decisions with ticker, horizon, probabilities, feature/SHAP vectors, base/raw score, ranked contributors, model/data references, and recorded risk tier. An explanation has text and optional JSON. Verification produces claim verdicts, four metrics and hard-gate failures. Emission adds fallback status and the rejected candidate/report.

## 5.7 Data Design

Historical artifacts remain the classifier source; no live quote, social observation or personalized classifier prediction is added. Context preserves missing values and rejects non-finite probabilities/attributions. The separate comparator fits population z-score statistics

on the first 148 saved raw feature rows of each corpus and scores only the later 38. This is a newly declared retrospective scaling run, not a reconstruction of the old CV baseline. Source hashes, fitting dates and statistics are recorded. No profile database is implemented.

## 5.8 Process / Behavior Design

The process validates a selected record, builds context, receives/renders a candidate, checks prose and JSON, applies hard gates, and accepts or verifies a fallback. The complete guided path additionally requires all feature values/contributions/directions/targets, family totals, meanings and raw-score scope in both representations. The local adapter bounds retries/timeouts and retains failed/truncated attempts. Actual inference now requires an original-publisher model and all weight-shard hashes to match the publisher pointers. The replacement evaluation is pending. A future constrained reference-reproduction result would not alone establish a comprehension benefit over the complete guided template.

The prototype profile refuses incomplete answers, scores capacity/tolerance separately, exposes matched/binding project caps, and reconciles the smaller score. The ability/willingness and time-horizon distinctions have external educational references [23, 24]; these do not validate project scoring. The separate comparator uses the profile score as R in C=R S+(1-R)T beside R=0.5. T averages five technical z-scores with inherited trend signs; S uses standardized headline polarity. Missing news contributes zero with disclosure; missing technical inputs or development sessions refuse output. Scores are not probabilities/returns. The mapping is unvalidated, and lower R does not guarantee a more conservative action. Saved classifier outputs remain unchanged.

An actual saved-feature semantics audit evaluates C(R)=T+R(S-T). Its slope is S-T, so increasing R changes the weight of the news score and does not impose a conservative/aggressive ordering. Across the 38 preserved historical held-out sessions per corpus, R=0 versus R=1 changes the comparator action on 25 Alpaca and 29 FNSPID rows. The guided UI now also offers the separate comparator on the new frozen and refit records; its default 21 December case shows Hold versus Sell under explicitly fictional example profiles. These are parameter scenarios on historical features, not observations of people or investment outcomes. The audit receipt retains the arithmetic and original source hashes.

# Chapter 6

# Iteration 1: Design and Implementation

## 6.1 Iteration Objectives

Preserve the existing project, validate the cached decision/explanation path, and make technical/news evidence inspectable. Establish the LLM input boundary and replay six recorded assistant-teacher examples; a live generator and measured user-comprehension benefit remain pending.

## 6.2 Selected Use Cases / Features

Select a historical corpus/session, inspect the decision/probabilities, read every feature's meaning and attribution, and inspect coverage/limitations. Fault checks reject corrupted confidence and show verified fallback.

## 6.3 Design

The frozen decision record remains the prediction source. A new builder combines its facts with reviewed definitions and implementation limitations. Streamlit and the standalone explorer reuse that context. Existing template verification remains the baseline.

## 6.4 Implementation

The active consolidated repository is on `main`; the earlier isolated branch is preserved in history. New code in `src/investify/explain/context.py` covers every feature, han-

dles missing values, rejects non-finite attributions/probabilities, and identifies unreviewed definitions. The Decision tab renders the context; the exporter uses the same records.

## 6.5 Completed Features / Module

The context, artifact tools, complete-guided verifier, report-retaining fallback, both live interfaces, prototype profile and separate retrospective risk comparator are implemented. Actual Streamlit widget tests pass. Six actual teacher explanations are saved and replayed in Teacher Study. Original social-candidate comparison now executes with recorded results in Chapter 7. Live LLM inference, social features in the saved predictor, personalized classifier predictions and remote GitHub execution remain pending.

## 6.6 Technical Implementation Details

Feature values and predicted-class SHAP contributions are separate. Family totals retain raw-score units. The verifier checks prose and JSON independently, including feature values and attribution targets. Displayed-text coverage, essential disclosures and hard contradiction gates prevent a correct payload from hiding an incorrect display. Strict mode rejects unsupported sentences. Evidence tools check schemas/context/templates/distributions/additivity and record artifact/runtime/source hashes.

## 6.7 Algorithms and Technical Decisions

Definitions follow actual features: dispersion is trailing variation of session-average polarity, and news attention is a headline-count z-score. Missing data stays missing. Indicator meaning and its use by this model are distinct [25, 26, 27].

## 6.8 Tools, Frameworks and Technologies

The portable path uses standard Python. The restored isolated viewer/test environment yields 212 selected passes and no skips; it is not the locked scientific environment. CI targets Python 3.11 on Windows/Linux with portable checks, frozen plans, comparator audits and a separate Streamlit AppTest job. Remote jobs have not run.

## 6.9 Demonstration Evidence

The continuation adds an explicitly labelled feature-level causality button: nine checks on synthetic fixtures detect four centred-window mutations, with no classifier fit. Scientific input/runtime absence disables the real re-run, and a later import/data failure is reported without a fabricated experiment result. Portable reports are invalidated after candidate/settings edits; record-loading errors retain selection/retry controls. These checks are distinct from browser visual QA. Artifact checks cover 186 records per corpus. The live HTTP exercise passes 62 explanation/risk cases. The separate comparator audit passes 228 arithmetic checks on actual held-out rows with labelled parameter scenarios. Browser visual QA remains pending. Both saved classifiers predict Buy throughout; no predictor-performance improvement is claimed.

## 6.10 Iteration Outcome

### 6.10.1 Versioned Scientific Corrections

The historical purger used calendar-day subtraction for observed-row labels. On the same labelled business-day fixture/test blocks it retains 18 training-label endpoints reaching the evaluation blocks; the corrected `observed-label-end-purge-v2` retains zero. Actual endpoints are calculated before feature filtering and carried to folds. Calibration candidates ending on/after first evaluation are excluded; a future-price shock leaves retained fixture inputs unchanged. Explicit frozen-classifier and refit-transfer modes now support a fair source-level comparison, with a genuine fresh Alpaca fitted comparison now recorded below.

Original scientific source/results are preserved. Cache joins refuse duplicate document keys or inconsistent probabilities, RSI edge windows have explicit behaviour, datetime units are normalized for the as-of join, and input ranges/ticker/revision/calendar contracts fail visibly. New outputs/cache directories are isolated. Full score/margin/truth arrays, calibration knots, scaler/model bytes and source/input identity are retained by the new trace mechanism. These earlier source/fixture repairs alone do not establish changed historical performance; the fresh-run findings below are separate. E_SAMEBAR_TEXT is a lag-rule ablation under the retained strict-prior join. The increment improves evidence coverage and transparency over the top-k numeric template and supplies a tested LLM input contract. It does not yet establish LLM usefulness, predictive skill, complete mid readiness, or deployment.

## 6.11 Recorded Assistant Teacher Construction Pilot

Under the user's latest direction, the current assistant supplies teacher examples before an alternative API is considered. The planned student remains Phi-3-mini; no additional Qwen teacher download is required. Six actual explanations were written from frozen historical NVDA records, and all six pass the current strict verifier. Input packets contain facts, reviewed definitions and controlled sentence patterns without a completed reference answer.

The supervision is hybrid: prose is actually assistant-generated, while the factual JSON payload is assembled deterministically from the original record. Both are checked, but the payload is not claimed as a teacher completion. Exact serving model ID/revision, decoding, token usage and per-example inference latency are unavailable and recorded as null.

Both corpora share date groups: 118 training dates, 30 validation dates and 38 preserved held-out test dates. The actual pilot has four training and two validation examples; test records are excluded. All saved source decisions are Buy, so neither class generalization nor adequate QLoRA data is established. Some held-out examples were previously explored; no first-ever untouched-test claim is made. Copied evidence replays original hash-bound raw persistence times and exact exports.

Original-student inference, QLoRA training, student quality and user understanding remain unmeasured. The inspected GPU has 10,240 MiB; the proposal's 16 GB envelope does not establish current-machine feasibility. Required training libraries, exact student checkpoint/tokenizer and measured token/VRAM budgets remain pending. Earlier Qwen-specific local pilot choices are inactive for the selected teacher/student workflow.

## 6.12 Fresh Alpaca Scientific Execution

On 12 October 2026 a separately identified, genuine Alpaca experiment executed the corrected pipeline. All 1,509 requested NYSE price sessions were present. The 6,768 revision-conservative news rows contain 5,748 distinct headlines, scored offline with original ProsusAI/finbert revision `0bc2ac10dda6929bcb1f77ccc43c7ec5c424f031`. The selected price basis is current-vintage SIP split adjustment, without cash dividend adjustment. News availability uses the later creation/update timestamp. Neither policy proves historical vendor availability. No FNSPID was downloaded; this run does not replace the preserved September results or reproduce their bytes.

Six arms at two horizons ran on shared row sets (1,309 one-day and 1,305 five-day rows), with six folds, fixed seed and unchanged model settings. The clean five-day macro-F1

is 0.2843 versus the training-distribution stratified-random baseline 0.3287, giving lift -0.0444 (95% fold-bootstrap interval [-0.0855,-0.0121]). The centred-Bollinger mutation gives five-day macro-F1 0.5390 and lift +0.2104 (interval [+0.1698,+0.2421]), demonstrating the deliberate canary. The one-day clean arm passes the configured control rule; the five-day clean arm remains below chance. E_SAMEBAR_TEXT remains a lag ablation.

| Calibration mode | Eval. rows | Macro-F1 | Brier before | Brier after |
|---|---|---|---|---|
| Frozen classifier | 186 | 0.2238 | 0.7929 | 0.6105 |
| Refit transfer | 186 | 0.2238 | 0.7760 | 0.6078 |

Table 6.1: Fresh revision-conservative Alpaca experiment, five-day held-out evaluation.

Five of 186 candidate calibration labels ended on or after evaluation start and were excluded. The 181 retained rows fit isotonic mappings; 186 evaluation sessions span 29 March–21 December 2023. Both current modes produce 186 Buy decisions. Before calibration, the frozen model predicts Buy 40, Hold 51 and Sell 95. Its fitted isotonic mappings make Buy largest on every evaluation row. Raw held-out macro-F1 is 0.3182 in frozen mode and 0.3116 in refit mode; calibration reduces both to 0.2238 despite improving Brier. A fixed class-prior comparator estimated from the 181 available calibration labels has Brier 0.6095 on the same evaluation rows, versus frozen 0.6105 and refit 0.6078. This descriptive comparison does not establish a useful learned forecast; its probabilities use no evaluation labels. Thus classifier refitting is not necessary for the observed fresh-run collapse. This is an observation about the current experiment, not proof of the historical cause. Lower Brier and all-Buy classifications must be reported together.

Actual inference from retained model bytes/scalers exactly reproduces the stored raw scores. Independent native XGBoost TreeSHAP matches every exported contribution and bias; the maximum frozen additivity residual is about $1.02 \times 10^{-6}$ against the unchanged $10^{-4}$ tolerance. Calibration, labels/endpoints and metric replay pass. Complete guided explanations verify on all 372 new records, separately from the historical 372.

The current source/input/runtime identities and execution logs are retained in `code/artifacts/scientific_runs/` and dated evidence receipts. The restored Python 3.11 runtime satisfies declared dependency ranges; it is not exact restoration of `uv.lock`. Engineering contracts record 179 passes and six legacy-input skips, with eight separate admission-refusal passes. No independently validated social accuracy, trading return, human benefit, student training or institutional approval is established.

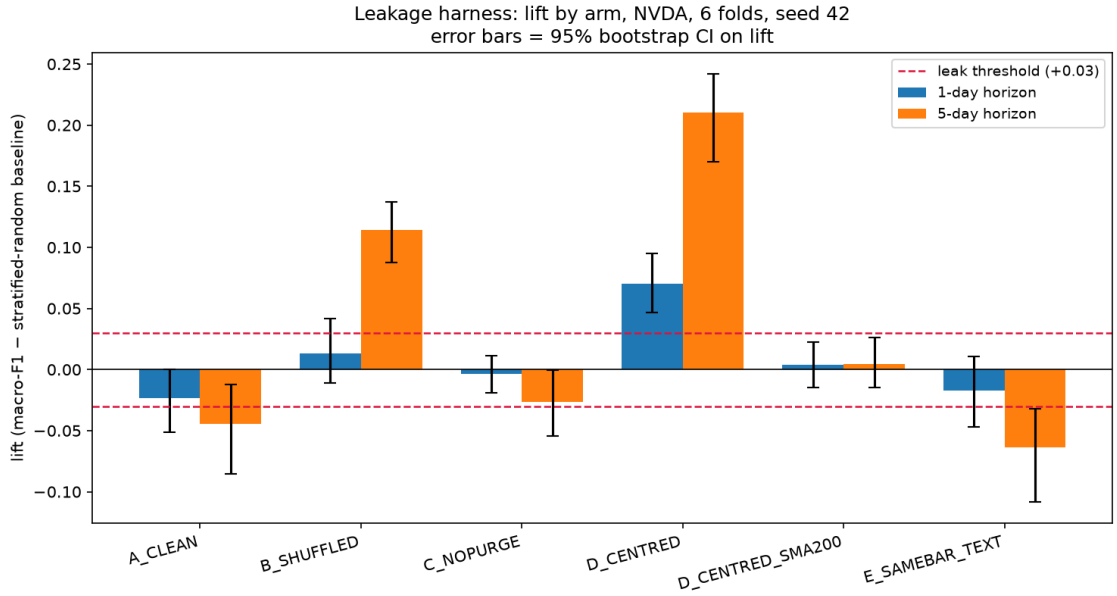



Figure 6.1: Actual fresh Alpaca leakage harness: lift relative to a training-distribution random baseline.

# Chapter 7

# Testing and Early Validation

## 7.1 Testing Strategy

Positive acceptance and labelled corruptions are both tested. Actual saved decisions support integration checks. Missing-input and corruption fixtures are test inputs, not synthetic substitutes for historical news or real LLM generations.

## 7.2 Test Levels and Techniques

Unit checks cover claim/schema validation, strict mode and fallback. Integration checks cover all 372 records. Local HTTP checks exercise the actual server and verification endpoint. Separate Streamlit AppTest cases exercise real widgets when its runtime is installed [28].

## 7.3 Test Cases

Important cases include correct JSON beside wrong displayed confidence, direction, value or class, correct prose beside wrong JSON, omitted essential claims, malformed/non-finite payloads, padding around a contradiction, unsupported sentences, and missing data/runtime. The appendix maps groups to requirements.

## 7.4 Test Results

Separate continuation core contracts have 153 passes with three shuffled-KFold cases excluded in the Python 3.11 viewer runtime. They use explicitly labelled fixtures, including

actual temporary Parquet writes and mocked model seams; they are not a full scientific or original-data suite. Six real-corpus loader cases remain skipped because their CSVs are absent. New V4 byte/receipt refusal and CPU tensor-serialization tests retain their own dated scopes; those earlier fixture receipts did not execute original weights; the genuine comparison below is separate. On 12 October 2026, the selected suite completed with **212 passed and zero skipped**, including 20 actual Streamlit AppTest cases. All 372 records retain schema/context/numeric-template and complete-guided-template acceptance. The live HTTP run passed 62 checks, including explanation and separate risk/refusal probes. The comparator audit passed 228 arithmetic/chronology/immutability checks on 76 actual held-out rows and three labelled profile scenarios. Runtime/source hashes and duration are in evidence JSON. This is not the complete scientific suite, a participant study or predictor-performance evaluation.

The default-mode fixture study has 112 cases: 80 corruptions rejected, 16 clean cases accepted, 16 vague-sentence cases accepted, and zero clean cases rejected. Vague-sentence acceptance remains a rate-threshold limitation. Strict mode rejects unsupported sentences but restricts accepted prose to the supported grammar.

## 7.5 Acceptance Testing / Validation

Programmatic acceptance is demonstrated for the explanation and prototype-comparator paths, including actual Streamlit widgets and HTTP health. Browser visual QA, participant comprehension and remote GitHub CI remain pending. No participant observations or member authorship are assumed.

The Social Study view's complete/partial flows are also exercised with clearly labelled benchmark/run fixtures: complete data exposes metrics and downloadable evidence; failed outcomes retain coverage/errors and refuse a paired comparison. Those widget tests do not establish actual sentiment-model quality.

## 7.6 Edge and Failure Cases

Correct JSON cannot hide empty text, an omitted decision or missing notice. Invalid claim structure selects verified fallback. Missing/zero feature evidence is preserved; non-finite authoritative probabilities or attributions are rejected. Missing records return errors; absent training inputs disable re-runs. The local adapter's unavailable-provider, malformed/truncated output and retry/fallback paths are tested with labelled fixtures. Incomplete profiles, development-date comparator requests and missing technical inputs are refused. Actual-model behaviour still requires real generations.

## 7.7 Defects and Corrective Actions

Additional regressions expose the calendar-day/session-label purge mismatch, calibration outcomes reaching evaluation, Pandas 3 join-key resolution skew, in-batch duplicate persistence, delimiter-hash ambiguity, start-only news-bound exclusion, implicit UTC assumptions, cache-key row expansion, RSI all-rising missing values and stale portable reports. Corrections preserve historical source and raw/frozen evidence. Current package and trace receipts must match their exact source snapshot; earlier passing XML is not proof for later edits. New regressions reproduced payload-over-prose acceptance, unchecked feature values/target class, payload-supplied display coverage, and dilution by repeated correct claims. Independent checking and hard gates correct these defects. The old test requiring correct JSON to override wrong prose was replaced with the corrected regression. Rejection reporting now retains the failed candidate's metrics instead of showing fallback success as the rejection reason.

## 7.8 Current Validation Status

Nine social-selection unit tests use explicitly generated fixtures. They check declared post/annotation provenance, exclusions, chronological freezing, code/data/model identity, initial-label agreement, full paired outputs and null metrics on failures. The actual AI reference pilot contains 200 real MSFT tweets: 117 usable labels (31 positive, 67 neutral, 19 negative) and 83 explicit exclusions. Its frozen split is 93 development/24 held out, with only two negative held-out references. One session pass is recorded with full prompt/input/raw-output/timing and declared model-identity provenance; agreement/kappa are null and independent human validation is absent. The later original-publisher paired comparison is recorded below. These tests are not sentiment-model study results, and schema validation cannot authenticate human independence.

Eight offline-inference tests use a labelled backend fixture. They check actual held-out text routing without human labels, declared preprocessing/class mapping, stable softmax, rejected incomplete/mismatched heads, unknown plans/models, missing weights and retained errors/truncation. Those early fixtures did not execute original weights; actual publisher-byte-bound Torch/Transformers inference now completes both candidates, as recorded below.

Python 3.11.9 and an isolated viewer environment were used, not the locked scientific environment. The viewer was restored from existing E: cached wheels plus two unchanged vendor pure-Python dependency sources/fallbacks, with provenance/licenses retained. Streamlit 1.63.0, NumPy 2.4.6 and Pandas 3.0.6 are frozen separately; dependency consistency passes. The original September raw/store inputs remain absent; a sep-

arately identified fresh Alpaca experiment now executes on restored genuine inputs. Local model probes returned no available identifier/completion; no LLM generation is claimed by those probes. The separate fresh science and original social-classifier runs retain actual execution evidence. The report is compiled as an unsigned working draft with layout inspection recorded separately; institutional review is incomplete. Both saved classifier folds remain constant-Buy.

## 7.9 AI Reference Pilot and Saved Calibration Audit

The actual AI reference pilot contains 200 real MSFT tweets: 117 usable labels (31 positive, 67 neutral, 19 negative) and 83 explicit exclusions. Its frozen split is 93 development/24 held out, with only two negative held-out references. One session pass is recorded with full prompt/input/raw-output/timing and declared model-identity provenance; agreement/kappa are null and independent human validation is absent. The later original-publisher paired comparison is recorded below. Ten additional annotation/import unit tests use labelled fixtures to check raw packet coverage, input/prompt tampering, single-pass null agreement, optional cross-checks and explicit exclusions. The actual raw-label batches are separate research artifacts. The Social Study widget also checks the AI-reference disclosure and missing-inference state.

The saved-artifact audit independently summarizes the preserved 372 historical records, all Buy; their historical cause remains unproven. The later genuine revision-conservative Alpaca experiment executes both corrected calibration modes on matching evaluation dates. Each still produces 186 Buy decisions. Actual retained-model, isotonic, label-endpoint, metric and native-TreeSHAP replay passes, and every new guided explanation verifies. The raw frozen classifier's diverse decisions collapse after its fitted isotonic mappings; refit transfer is therefore not required for the fresh observation. Five unavailable calibration targets are excluded. The new experiment's clean five-day score is below its random baseline, while the deliberate centred feature canary fires. Chapter 6 records exact metrics, input/runtime limits and the distinction from September results.

The frozen-reference evaluability audit selects neutral as the development-majority comparator, obtaining 15/24 exact held-out AI-reference agreement and macro-F1 0.256410, with zero positive/negative recall. This deterministic comparator is not candidate inference or independently human-validated accuracy. One held-out prediction changes exact agreement by 4.17 percentage points; negative recall moves in 50-point steps. No pair meets the declared token-Jaccard 0.8/minimum five-token heuristic, which does not establish independence or exclude shared events/pretraining overlap. Reference and split hashes are unchanged. The separate generative-model replacement remains pending verified weights and available access; the original-classifier study below is complete.

## 7.10 Teacher Pilot Evidence

The six recorded examples, four-training/two-validation split, serving-identity limitations and untrained Phi-3-mini student are documented once in Section 6.11. No training efficacy or human-comprehension result is inferred.

## 7.11 Executed Original-Publisher Social Comparison

On 13 October 2026 both pinned original classifiers completed the same 24 chronologically held-out MSFT posts. The frozen reference contains 7 positive, 15 neutral and 2 negative AI labels. Source posts, reference labels, exclusions, candidate identities, preprocessing and the original V3 split are unchanged.

| Candidate | Macro-F1 | Exact agreement | Positive recall | Negative recall |
|---|---|---|---|---|
| CryptoBERT | 0.2564 | 15/24 | 0/7 | 0/2 |
| Cardiff | 0.6016 | 16/24 | 4/7 | 1/2 |

Table 7.1: Agreement with frozen AI reference labels; not independently human-validated accuracy.

CryptoBERT predicts neutral on all 24 posts, matching the development-majority neutral comparator. Cardiff is the exploratory benchmark leader. A single prediction changes exact agreement by 4.17 percentage points, and negative recall changes in 50-point steps. Public-post pretraining overlap and shared events remain unexcluded. These results do not establish NVDA transfer, return prediction, significance, suitability or deployment readiness. No predeclared adequacy threshold exists in the retained plan, and the deployment selection remains unset pending actual criteria and review.

Both original publisher revisions, weight hashes and all loading-affecting tokenizer/config files are verified. Original Cardiff BIN weights were losslessly serialized: all 204 tensor keys, dtypes, shapes and raw bytes match. Its two unused native pooler parameters are retained during modern loading, with exact classifier-logit parity verified. No classifier head is missing, mismatched, substituted or randomly initialized. Modern set-valued loader diagnostics are normalized for recording; nonempty diagnostics still refuse inference.

The first Cardiff loading refusal is retained. Both models reran under a new source-bound plan, retaining per-post raw logits, processed-text hashes, token counts, truncation, runtime identity and probability outputs. Independent receipt/logit comparison reproduces the reported table; repeated CryptoBERT outputs are identical. The Social Study view displays recomputed recorded scores with AI-reference limitations and download evidence. The earlier recorded-social scope has 25 viewer/selector passes, including 21 Streamlit

cases; the later guided-demo scope has 28 passes, including 24 AppTest cases. These are overlapping suites; 37 current loader/pooler checks and nine subtests pass on their declared scopes. These scopes overlap earlier evidence and are not summed as a full-suite total.

# Chapter 8

# Project Management and Development Evidence

## 8.1 Work Completed

The isolated copy preserves original history/documents. Completed work includes complete feature/family evidence, independent prose/JSON checks, strict mode, rejection/fallback reporting, local LLM transport/evaluation plumbing, both live interfaces, restored viewer runtime, authoritative-record validation, an explicit prototype profile and a separate retrospective risk comparator. The selected suite has 212 passes and no skips; local evidence covers 372 records, 62 live HTTP cases and 228 comparator arithmetic probes. CI jobs are prepared.

## 8.2 Work Remaining

The real fresh Alpaca scientific run and original-publisher paired social comparison now complete, with retained raw execution, source/input identity and replay evidence. Cardiff leads the small AI-reference benchmark; final adequacy criteria, independent validation and deployment selection remain open. Source/raw labels, exclusions and the original V3 chronology are preserved.

Current gaps are institutional report/deck inspection and completion, genuine member contributions and review/rehearsal, current repository/remote CI evidence, live LLM usefulness and agreed risk semantics. The original September bytes and exact scientific lock are not reproduced. Student inference/training and broader acceptance work remain later milestones.

## 8.3 Contribution of Each Team Member

The team must record actual implementation, review, testing and defence preparation. No member-specific completion table is inferred from assumed allocations.

## 8.4 Repository and Version Control Evidence

The consolidated repository is on `main`. Original baseline `1dc24e0` and its isolated branch are preserved in history; local commit `93d0d60` records the fresh scientific increment. Current verification evidence records source/data hashes. Upstream access remains unverified; local workflow configuration is not evidence of a successful remote CI run.

## 8.5 Milestones and Commit Evidence

The checkpoint distinguishes the initial context increment from verifier/local-demo work. Historical experiments are preserved separately from current test/HTTP evidence. Reproduction instructions appear in the repository appendix.

## 8.6 Tools and Tool-Assisted Development

The coding assistant contributed code, regression tests, drafts and programmatic validation. Earlier handoffs declare GPT-6.1 Sol/Xhigh as selected session configuration; the exact serving revision is not provider-authenticated. Student review, source verification, institutional declarations and accurate tool disclosure remain necessary. Student authorship is not inferred from assistant work.

## 8.7 External Components and Dependencies

The baseline uses FinBERT, XGBoost, isotonic calibration, TreeSHAP and Streamlit. The portable demo uses standard Python HTTP/JSON facilities and existing record/verifier code. The retained social labels were generated by the assistant in one actual session pass and disclosed as AI references. Six recorded assistant-teacher explanation completions are retained separately. No API/local student inference or human annotation is claimed.

## 8.8 Risks and Mitigation

Both interfaces expose recorded evidence and failure cases. The restored scientific runtime executes a separate fresh snapshot; original-byte and exact-lock limitations remain disclosed. Unknown deadline/owners affect scheduling; unavailable model access prevents measuring LLM usefulness. Constant-Buy classifier output prevents a predictive-skill claim. Prototype profile-to-blend assumptions require review and do not establish suitability.

## 8.9 Security and Privacy Considerations

The local demo serves historical decisions on loopback and has no trading/account access. Profile persistence, production authentication and deployment security are beyond this increment.

## 8.10 Current Limitations

Browser visual QA, GitHub CI, real LLM evaluation, personalised classifier modelling and deployment remain pending. The new science and original social-classifier comparison are executed and recorded separately. The separate risk comparator is a tested prototype, not validated investor advice. Institutional metadata, signatures, remaining research and rendered-deck review are incomplete. The working report compiles; its dated delivery receipt records PDF layout inspection separately.

## 8.11 Teacher Pilot Evidence

The six recorded examples, four-training/two-validation split, serving-identity limitations and untrained Phi-3-mini student are documented once in Section 6.11. No training efficacy or human-comprehension result is inferred.

# Chapter 9

# Remaining FYP Plan

## 9.1 Remaining Scope

The full approved advisory scope exceeds the current historical news-only spike. Remaining work includes social transfer/divergence, genuine risk conditioning, multi-equity evaluation, usable calibration/decision behaviour, serving parity, fault handling, local language-model evaluation/adaptation, deployment and participant validation. Transaction execution remains excluded.

## 9.2 Remaining Modules and Features

Restore the full runtime/data, establish the original Phi-3-mini student checkpoint, evaluate raw and emitted generations, and reconcile capacity/tolerance/lambda specifications. The paired original-candidate inference on 24 frozen held-out posts is complete; agree adequacy criteria and review the result before deployment selection. The 117 usable references from 200 acquired posts remain explicitly AI-generated. Independent human review remains optional and unmeasured. Use the declared news-only branch only with an accurately documented comparison/deviation decision; unavailable inference does not establish that neither candidate transfers. Extend beyond NVDA after acquisition/alignment assumptions and evidence are reproducible.

## 9.3 Remaining Iterations

The next increment validates model/runtime access and the constrained pilot. Subsequent increments integrate risk and degraded operation, social evidence/divergence and broader model evaluation, then serving/deployment and participant acceptance. QLoRA

and teacher/student work retain their later schedule; the panel feedback does not automatically make them mid deliverables.

## 9.4 Implementation Plan

Use the frozen explainer selection and deterministic comparators before modifying its prompt on evaluation records. Record checkpoints, model identity, data hashes and failed attempts. Restore raw/store inputs through documented acquisition or approved copies. Team-agreed owners and actual review/implementation contributions must be recorded rather than inferred from historical allocations.

## 9.5 Testing and Validation Plan

Run the scientific suite with documented inputs and locked dependencies, execute actual Streamlit/browser checks and GitHub CI, and assess prediction/calibration together with class distribution. For explainers, report first-attempt errors, retry/fallback rates, supported-claim/definition coverage, and human comprehension separately. The social pilot uses explicitly declared AI reference labels with prompt/input/raw-output/run provenance; candidate scores measure agreement with those references. Independent human validation is unmeasured, and optional independent review can strengthen it. Later validation covers applicable runtime requirements, fault behaviour, serving parity and participant tasks.

## 9.6 Deployment Plan

A delivery target remains unselected. Prepare a reproducible service and installation/deployment evidence after clean-host validation. The loopback demo and a workflow file alone do not constitute production deployment or successful continuous delivery.

## 9.7 Tentative Timeline

No assessment deadline has been confirmed. The sequence is dependency-based: access/restoration; explainer pilot; risk/social integration; broader evaluation; serving/deployment; acceptance and handover. Calendar dates and student ownership require team/supervisor agreement. The initial 45–75 person-hour mid-package estimate included completed work and is not a new remaining-time estimate.

## 9.8 Expected Final Deliverables

Reproducible source and data acquisition, model and evaluation evidence, audited decision and explanation paths, tested interfaces and degraded states, deployment/handover instructions, actual contribution/meeting records, final report/presentation and participant acceptance evidence. The staleness-gate design remains a later FYP-2 item unless scope is explicitly changed.

# Chapter 10

# Conclusion

Investify has a substantive historical pipeline and a tested explanation increment. Complete technical/news evidence display, independent prose/JSON checking, hard contradiction rejection and verified fallback improve the proposal-stage demonstration. The local LLM adapter and frozen pilot protocol are implemented with explicit no-model behaviour, exact maintained definitions and fair deterministic comparators. They do not establish the quality or usefulness of an unconnected model.

The preserved historical evidence remains separate from completed fresh runs. A real Alpaca experiment executes all six leakage arms at both horizons, both corrected calibration modes and retained-model/native-TreeSHAP replay. The deliberately centred canary fires, but the clean five-day score is below its random comparator and both calibrated modes remain all Buy. Lower Brier than the raw model does not establish useful decisions or superiority to a simple class-prior forecast.

Original CryptoBERT and Cardiff each complete 24 held-out MSFT posts, with macro-F1 agreement 0.2564 and 0.6016 against frozen AI references. Cardiff is the exploratory benchmark leader; the small reference, two negative examples, public pretraining overlap and lack of independent human validation prevent a broad accuracy or deployment claim. The recorded Social Study viewer validates both raw output/receipt bundles. Final selection criteria remain unset.

Continuation should complete institutional report/deck inspection, actual metadata and signatures, member contributions and genuine review/rehearsal, while preserving weak findings and original evidence. Live LLM usefulness, student adaptation, broader generalisation, risk suitability and deployment remain unmeasured. No historical ownership or participant evidence is manufactured.

The current assistant teacher construction pilot records six actual verified explanations with four training and two validation examples; the planned Phi-3-mini student remains untrained. Exact teacher serving revision is unavailable, factual JSON supervision is

assembled from original records, and no fine-tuning efficacy or human-understanding result is claimed.

# Appendix A

# Proposal Action Register

The exact supplied panel wording and the original proposed response register are preserved in the source snapshot. The table below records current working evidence; it does not declare supervisor-approved closure or invent owners/deadlines.

| ID | Requested clarification | Current evidence and remaining closure |
|---|---|---|
| PF-01 | Apparent contribution | Problem/contribution and research comparison drafted. Novelty/closest-work assessment and team review pending. |
| PF-02 | LLM's user benefit | Adapter, retries, fallback and comparators implemented; six actual teacher examples replayed. Live inference, added explanatory content and comprehension benefit remain unestablished. |
| PF-03 | Technical/sentiment explanation | All feature context and exact maintained definitions checked. No social observations or unrestricted semantic verification claimed. |
| PF-04 | Refined problem statement | Revised wording in introduction and contribution material. Team/supervisor agreement pending. |
| PF-05 | Pretrained models | Inventory separates FinBERT, candidate social classifiers, the current assistant teacher and planned Phi-3-mini student. Exact teacher serving revision is unavailable; exact student checkpoint and student inference remain pending. |

| | | |
|---|---|---|
| PF-06 | Explicit metrics | Prediction, calibration, explanation, social, user and system matrix and current scoped checks. Unavailable measurements labelled. |
| PF-07 | Improved PoC beyond leakage | Corrected verifier, live local interface, corruption and fallback checks, and versioned evidence. Supervisor/browser demonstration pending. |
| PF-08 | User transparency | Historical/constant-Buy/social/risk limits, rejected reports and labelled fallback. Full journey and human comprehension pending. |
| PF-09 | Stock-market understanding | Referenced domain/research notes and project design implications. Corporate-action/macro/regime study and member defence review remain. |

# Appendix B

# Detailed Requirements

| Requirement | Component | Evidence |
|---|---|---|
| MID-FR-01 | Schema/catalogue/input handling | Actual-record and missing/invalid-input checks |
| MID-FR-02 | Context and both interfaces | All 372 records; complete coverage tests |
| MID-FR-03 | Verifier | Prose/payload contradiction regressions |
| MID-FR-04 | Verifier/emission | Hard gates, omissions, malformed claims, verified fallback |
| MID-FR-05 | Emission/reporting | Retained failed report and HTTP assertions |
| MID-FR-06 | Strict prose gate | Clean acceptance and unsupported-sentence rejection |
| MID-FR-07 | Structural readiness/UI control | Missing, empty, truncated, invalid-type and read-denied fixtures; module discovery and actual widget checks |
| MID-FR-08 | Local adapter and interfaces | Retry, unavailable-provider, malformed/truncated output, loopback transport and fallback tests |
| MID-FR-09 | Guided template and verifier | All 372 guided records; modified-definition and coverage checks |
| MID-FR-10 | Evaluation command | Disjoint chronological selection, plan-only and zero-completion tests |

| | | |
|---|---|---|
| MID-FR-11 | Complete guided contract | Non-top-k, missing/zero, family, target, scope and two-source coverage tests |
| MID-FR-12 | Prototype profile/form | All 32,400 answer combinations; cap, incomplete-answer and actual widget tests |
| MID-FR-13 | Retrospective scaling | Both 148/38 splits, held-out perturbation and development/missing-input refusal |
| MID-FR-14 | Separate comparator | Actual HTTP/AppTest independence; 228 real-row/labelled-scenario arithmetic checks |
| MID-FR-15 | Social-study provenance/plan | Fixture provenance checks plus the actual frozen 200-post/117-reference AI study |
| MID-FR-16 | Sentiment comparison tool | Matched-plan, per-model coverage/refusal, null partial metrics and explicit ties/raw-output tests; actual paired 24-post scores recorded |
| MID-FR-17 | Offline classifier runner | Eight backend-fixture checks; head/mapping/weight refusal, no label inputs, raw/error/truncation retention; both pinned original classifiers executed |
| MID-FR-18 | Recorded teacher import/export | Seven fixture tests plus actual evidence replay and viewer checks; six real examples, hybrid target disclosure and separate exports |

Mapping to the complete proposal baseline and final acceptance study remains pending. Exact functions and named passing tests are retained in `code/docs/requirements_traceability.json`; `tools/verify_requirements_traceability.py` checks their existence and matching selected-suite execution. The working Markdown matrix also distinguishes each original first-mid commitment from downstream saved-record engineering evidence.

# Appendix C

# Detailed Test Cases

The table records the earlier 12 October selected engineering scope. Later fresh science, original-classifier and guided-demo receipts below supersede its pending states; scopes overlap and are not summed.

| Group | Scope | Status |
|---|---|---|
| Existing verifier/emission | 20 checks, including clean fixtures and corruptions | Passed |
| Context | 5 tests; actual records, absent/missing features | Passed |
| Verifier regressions | 11 tests; contradictions, omissions, schema, padding, strict mode, report retention | Passed |
| Portable HTTP | 11 tests; verification/fallback, invalid inputs, profile contract and independent risk comparator | Passed |
| Portable request ownership | 12 controlled Node cases execute real inline handlers with limited DOM/deferred-fetch fixtures; separate eight pure-state assertions | Passed; browser QA pending |
| Harness structural readiness | 10 tests; missing/empty/truncated/wrong-type/read-denied files, module discovery and explicit preflight limits | Passed |
| Local generator | 12 tests; definitions, positive acceptance, retries/fallback, malformed/truncated output, configuration and provenance | Passed |
| Provider transport | 2 actual loopback HTTP fixture checks | Passed |

| | | |
|---|---|---|
| Evaluation protocol | 7 checks; chronological split, frozen plan enforcement, no-inference/zero-output accounting | Passed |
| Complete guided evidence | 8 tests; all-feature coverage, non-top-k corruptions, missing/zero, family and scope checks | Passed |
| Risk profile/comparator | 10 tests; 32,400 answer combinations, chronology, scaling isolation, failure and classifier independence | Passed |
| Authoritative record validation | 4 tests; non-finite fields, vector/rank/value consistency and invalid identity/horizon | Passed |
| Social-selection tooling | 9 labelled-fixture tests; human-label provenance schema, adjudication, fixed split, paired coverage and metric/refusal paths | Passed; actual study pending |
| Source revision/handover | 5 tests; checked Git-free inventory, explicit working-tree status, modified/unsafe inventory refusal and ancestor-repository isolation | Passed |
| Offline sentiment inference | 8 labelled-backend tests; frozen identity, text/label mapping, head refusal, failures, truncation and absent weights | Passed; actual model pending |
| AI annotation/import | 10 labelled-fixture tests; blinded packets, retained initial labels, null single-pass reliability, input/provenance/persistence checks | Passed; human validation pending |
| Generative sentiment protocol | 6 labelled-provider tests; frozen parent split/runtime, raw failures and identity/source-drift refusal | Passed; actual inference pending |
| Original-publisher gates | 4 labelled-file tests; exact catalog identity, complete shard hashes and receipt replay | Passed; verified real weights pending |
| Recorded teacher import/export | 7 fixture tests; date grouping, actual-versus-fixture gate, independent prose checks and separate exports | Passed; student pending |

| | | |
|---|---|---|
| Streamlit AppTest | 20 tests; existing flow plus explicit profile, real comparator and stale/development refusal | Passed |
| Live endpoint exercise | 62 checks, including explanation cases and separate profile/refusal probes | Passed |
| Comparator arithmetic audit | 228 checks on 76 held-out real rows and three labelled parameter scenarios | Passed |

The selected suite totals 212 passes and zero skips. Test XML, fixture confusion matrix, artifact/source validation, comparator audit and live HTTP evidence are in `Investify/evidence`. Streamlit cases ran in the isolated restored viewer environment; remote CI has not run. Scenario profiles are deterministic probes, not participant observations. The dated initial continuation receipt is `background_readiness_tests_20261012.xml`; the later 188-pass receipt is `background_delivery_tests_20261012_v2.xml`. The final current receipt and source bundle identify any subsequent matching snapshot. The separate Node handler receipt is `background_portable_handlers_20261012_parallel.json`; its twelve cases are not added to Python-suite totals and perform no live server/model/browser requests. The 18 requirement IDs have exact source/test anchors checked by the traceability tool. This establishes execution of named regressions, not complete scientific or institutional acceptance.

The matching post-recovery receipts are `recovery_selected_tests_20261012.xml` (212 passes, twenty Streamlit cases) and `recovery_core_tests_20261012.xml` (153 scoped passes, three deselected). These overlap and are not added together. Separate read-only Python 3.13 probes execute three unchanged shuffled-split function bodies with real sklearn and replay actual isotonic helper fits on explicitly labelled numeric fixtures. Neither probe restores the original scientific lock or uses historical market inputs.

The 13 October guided-demo receipt records 28 passes and zero skips, including 24 AppTest cases and four recorded-study selector checks. It verifies fresh defaults, explicit fictional profiles, profile reset, navigation state and failure/fallback. The 179-pass scientific contract scope retains six legacy-input skips, with eight separate admission-refusal passes. Both original classifiers complete the same 24 held-out AI references. Chapters 6 and 7 document these actual studies; browser review, human validation and remote CI remain unexecuted.

# Appendix D

# Repository and Reproduction

## D.1 Active Workspace

Work in `E:/Uni_Stuff/FYP/Investify`; run Python commands from its `code` directory. The local branch is `main`, with no configured remote. The original checkout and source archives remain separate. Local commit `93d0d60` records the fresh scientific checkpoint; subsequent source and artifact changes retain dated receipts.

## D.2 Guided Demo and Historical-Record Demonstration

The current Streamlit app has a Start here route, fresh corrected results as defaults and three decision stages. Run from the code directory:

```
python -B -m streamlit run app/streamlit_app.py ‘
  --server.address=127.0.0.1 --server.port=8507 ‘
  --server.headless=true
```

Open the printed loopback URL. Fictional profile examples are explicit user actions; no participant data is assumed. The original portable interface remains available separately below. Use the restored viewer environment or the code's portable Python path:

```
python -B scripts/serve_mideval_demo.py --port 8768
```

Open the printed loopback address. The demonstration validates historical records and explanations; it does not fetch prices, retrain models or generate unconfigured LLM output.

## D.3 Executed Fresh Scientific Snapshot

The separately identified snapshot lives under `code/artifacts/scientific_runs/alpaca_revision_conservative_20261012_run01`. It retains configuration, a new price store, corpus manifest, all-arm results, both calibration-mode records/traces and execution logs. FinBERT scores are cached on E: under the same run identity. Original September bytes and the exact scientific lock are not reproduced.

Use `code/scripts/prepare_fresh_alpaca_science.py` for new receipt-verified input preparation, `code/scripts/prepare_scientific_recovery.py` for preflight/execution and `code/scripts/verify_fresh_science_run.py` for actual model/calibration/native-TreeSHAP replay. Each new execution must use new output/cache/receipt identities. The current scientific Python 3.11 environment is separate from the historical viewer lock.

## D.4 Original Social-Classifiers

The original V3 benchmark, label freeze and chronology remain unchanged. `code/artifacts/mid_eval/social_study_20261013_run02` contains the new source-bound plan, both original-classifier bundles and their recomputed comparison. Its complete publisher catalog is retained in the preceding preparation directory. Weights and the disclosed lossless Cardiff serialization remain on E: caches. The recorded viewer uses `social_latest_study.json`; it recomputes receipt-bound agreement scores rather than inferring new predictions.

## D.5 Report and Handover Limits

The supplied institutional class and logo retain their original bytes. Main-source compatibility settings allow the portable Tectonic engine to compile the draft without changing that class. Compiler/cache/build identities and layout checks are recorded separately. Required institutional facts, applicable signatures and actual team review remain open. Existing ZIPs and receipts prove their own dated snapshots; no remote CI, deployment, student ownership or final submission is inferred from local checks.

# Appendix E

# Additional Design Diagrams

The implemented architecture appears in Chapter 5. The following diagram describes the current runtime/JSON objects, not a deployed relational database. Saved records are inputs; explanations, verification and profile-comparator results are transient. A full-product database design remains pending.

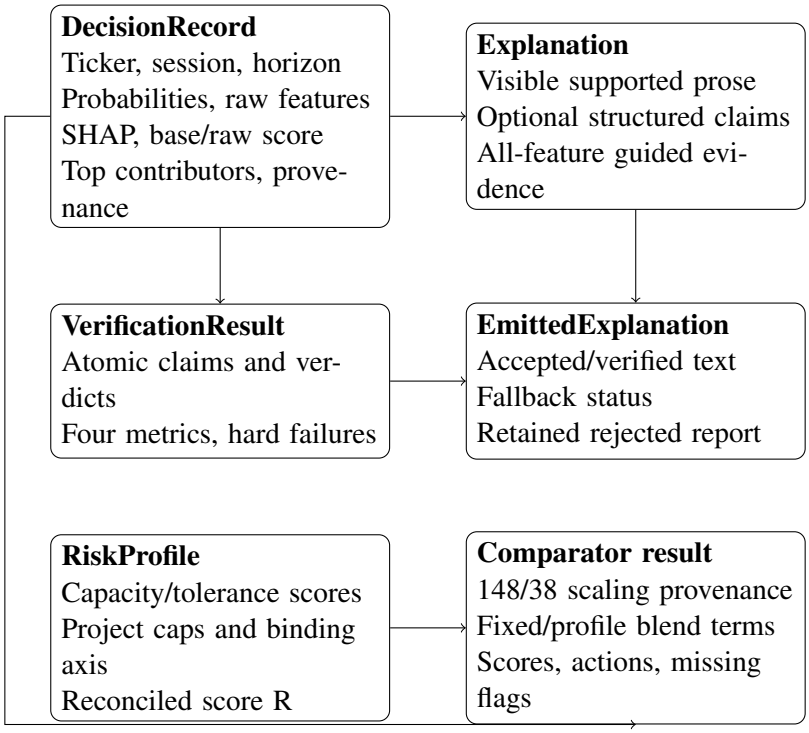



Figure E.1: Implemented record, explanation and profile-comparator object relationships.

# Appendix F

# Meeting and Progress Evidence

Actual meeting dates, participants, supervisor feedback and agreed actions have not been supplied for this increment. No attendance, approval or member contribution is inferred. The team must add authentic records before submission, keeping dates/participants/actions traceable to their original evidence.

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